

Peoplesoft ID:



EEE3094S 2024: Exam - Memo

- The exam is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your peoplesoft ID.
- There are five questions totalling 103 marks, but the maximum mark is 100.
- You have 15 minutes reading time and 3 hours to complete the exam.
- The last page is a formula sheet.

Question 1 [9 marks]

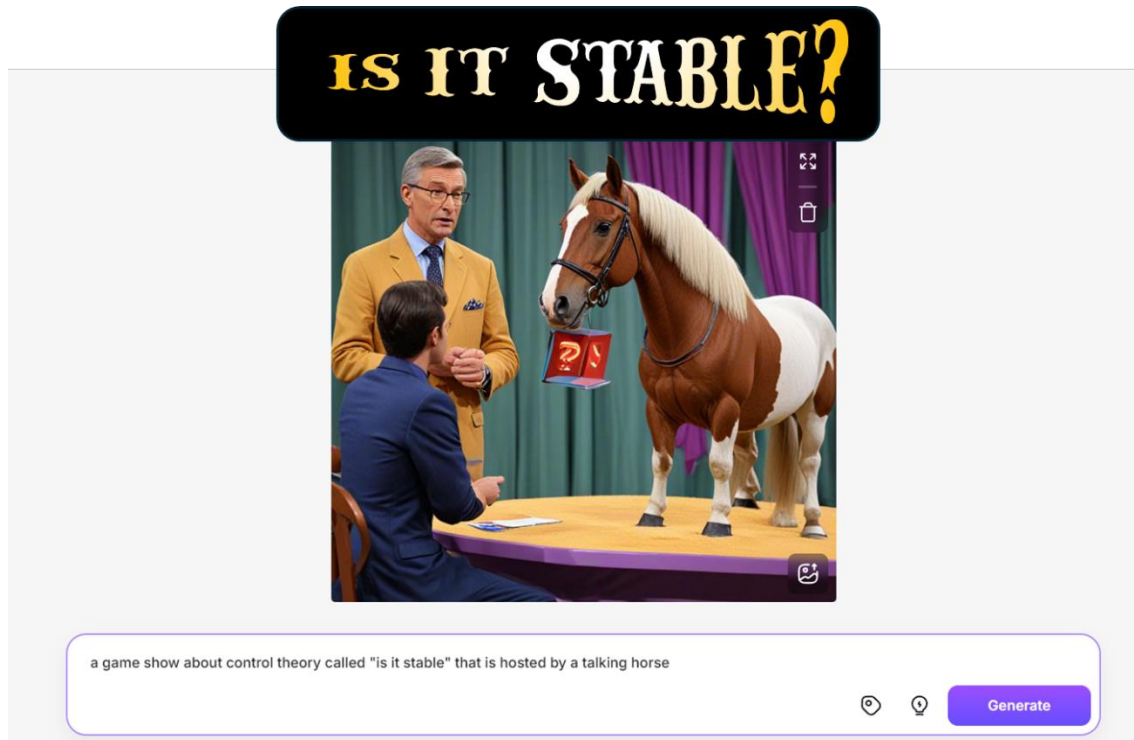
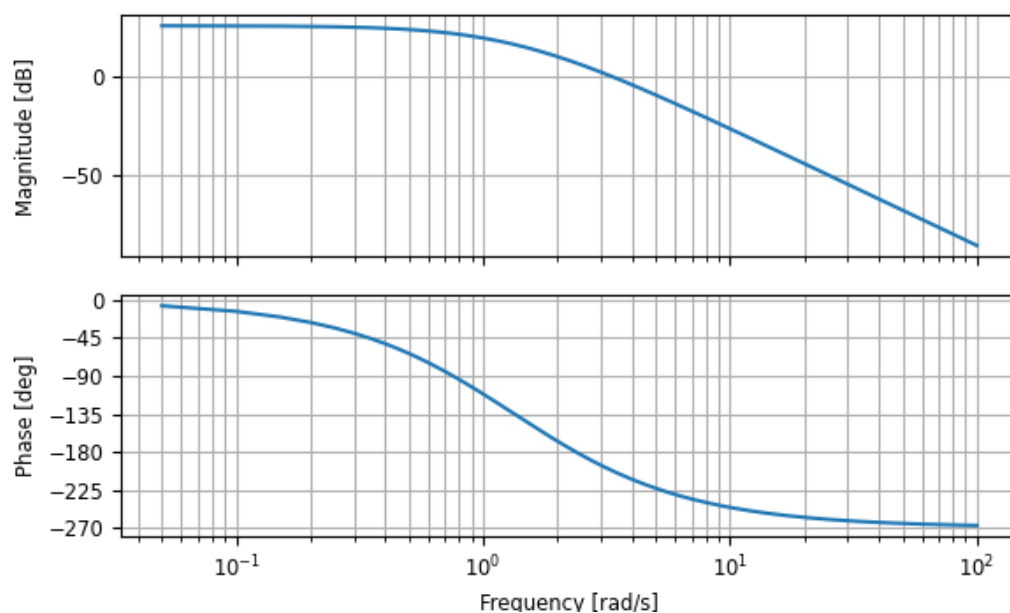


Figure 1: You know what, Midjourney AI? I'll take it.

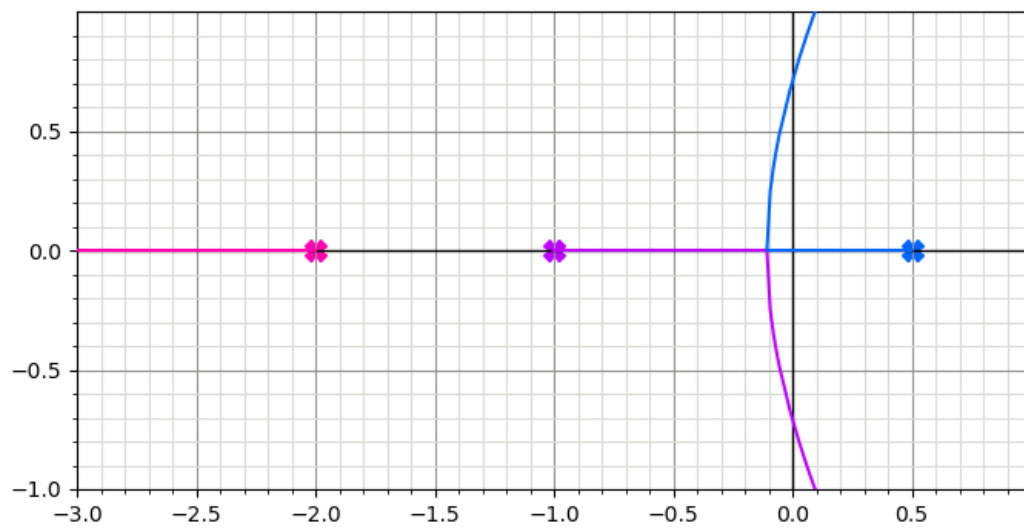
You've heard of Netflix's *Is it Cake?* But what about *Is It Stable?* The game show with way more feedback control theory, and way fewer tasty treats. Here's how it works: you, the contestant must take a look at each of our three eligible systems, and tell us "is it stable?"

Possible answers might be: Yeehaw! (yes) Neigh! (no) Or, it depends! But you need to say what it depends on. For every system you correctly call (provided you give a sensible explanation), you will win 3 marks!

- The closed-loop system, if the open-loop frequency response encircles -1 on the Nyquist plot twice.
- The system with this open-loop frequency response, when put in closed loop:



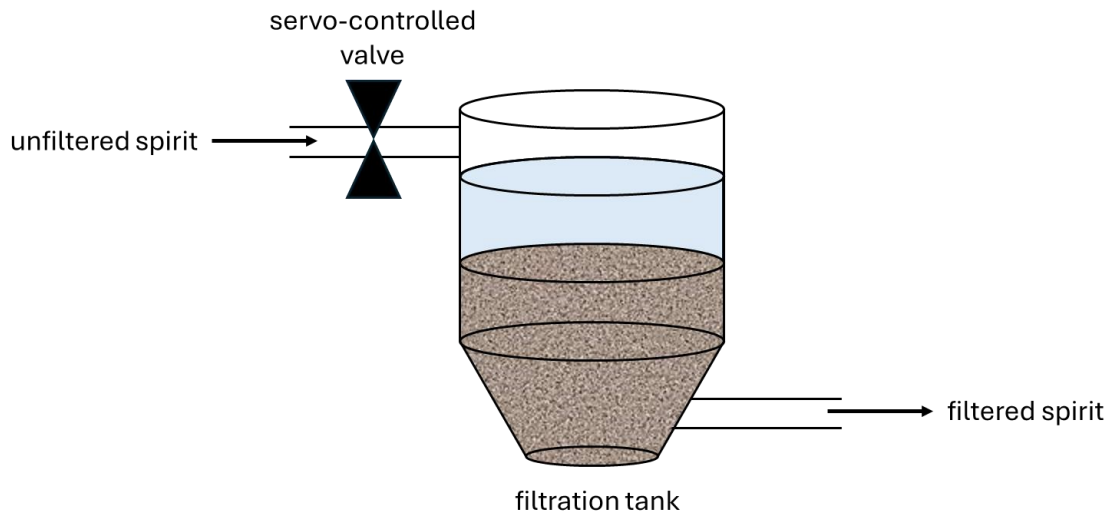
c) The system with this root locus, when put in closed loop with a proportional controller:



Solutions:

- a) It depends on the direction of the encirclements and whether the open-loop system has any unstable poles. The Nyquist stability criterion requires the number of *clockwise* encirclements, and the number of open-loop poles in the right-half plane. [3 marks]
- b) Neigh! The system has a gain exceeding 1 (0 dB) at the phase crossover frequency, so it is unstable in closed loop. [3 marks]
- c) It depends on the gain of the proportional controller. The open-loop system is unstable, but there are values of gain where all three poles are in the left-half plane. [3 marks]

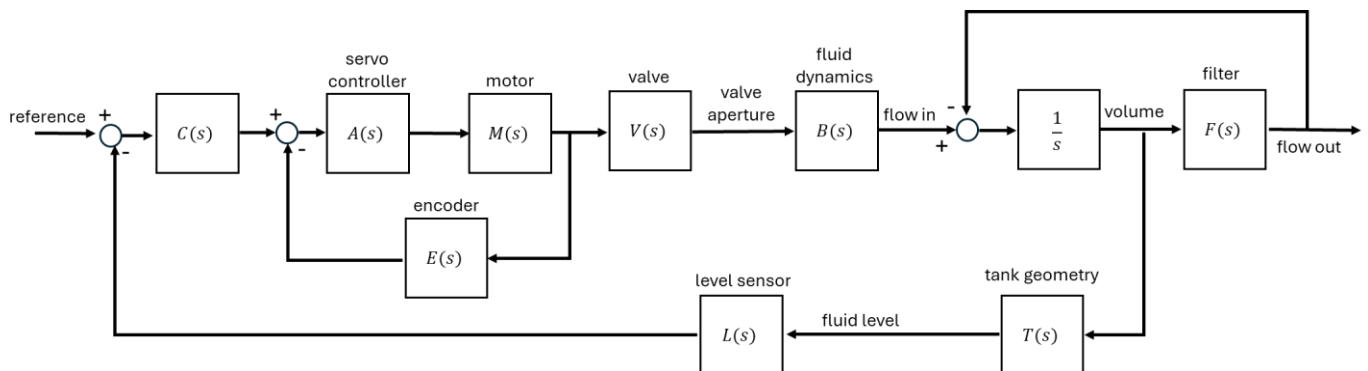
Question 2 [12 marks]



Your friend got into distilling during the quarantine, and this has now turned into a thriving craft spirits business. Believing that you know something about control engineering, they have asked for your help automating their process.

Specifically, they want you to help control the fluid level in their vertical filter. This is a tank containing a carbon filter that the unfiltered spirit is fed into via a variable valve controlled by a servo. Liquid then flows out of the tank at a rate determined by the level in the tank and the properties of the filter. A level sensor monitors the height of the liquid in the tank and feeds it back to the microprocessor controlling the input servo.

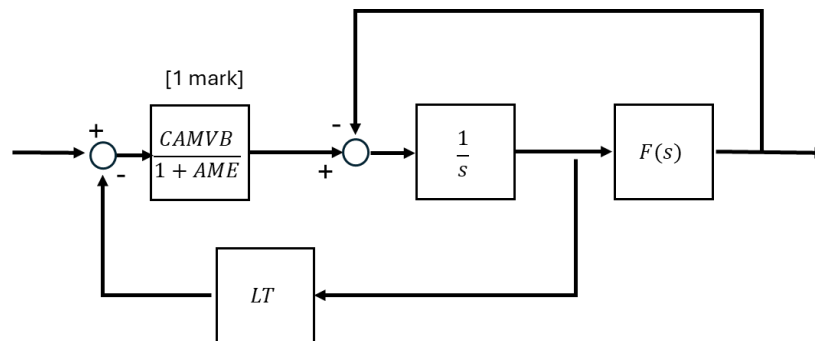
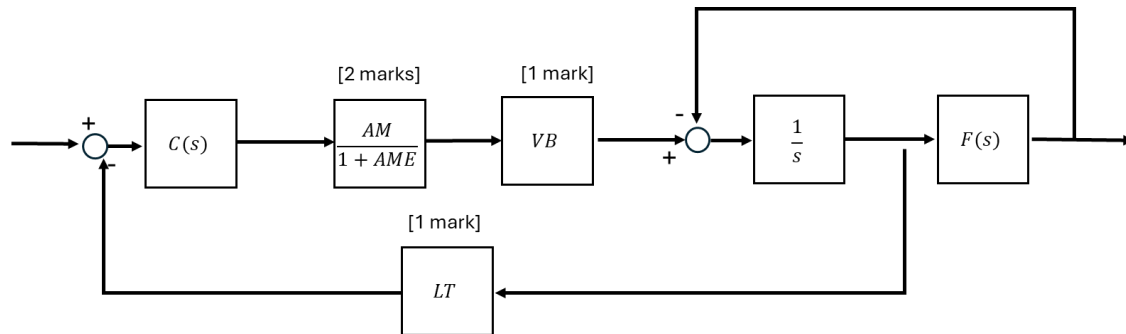
A block diagram of this system is shown below:



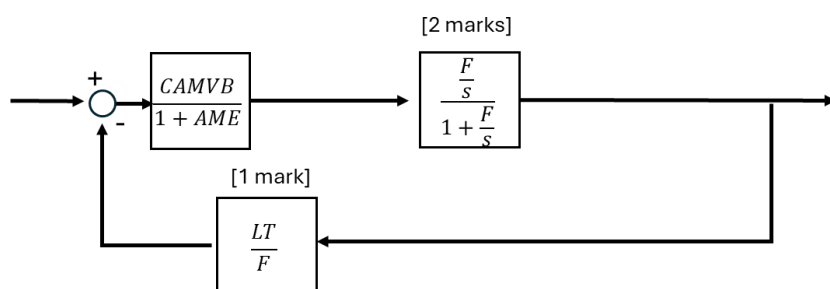
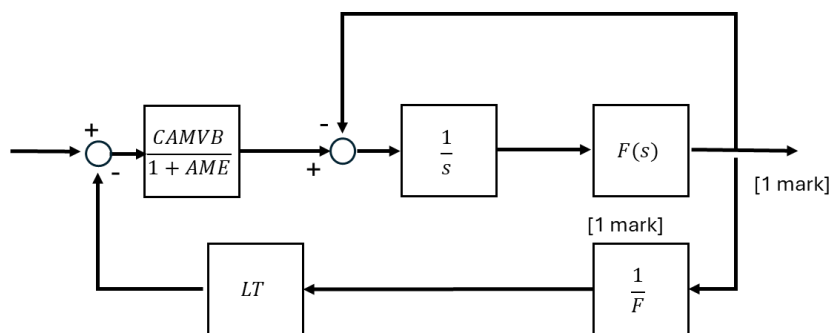
Simplify the block diagram into a single transfer function. [12 marks]

Solution:

Simplify the cascaded systems and simple loops (5):



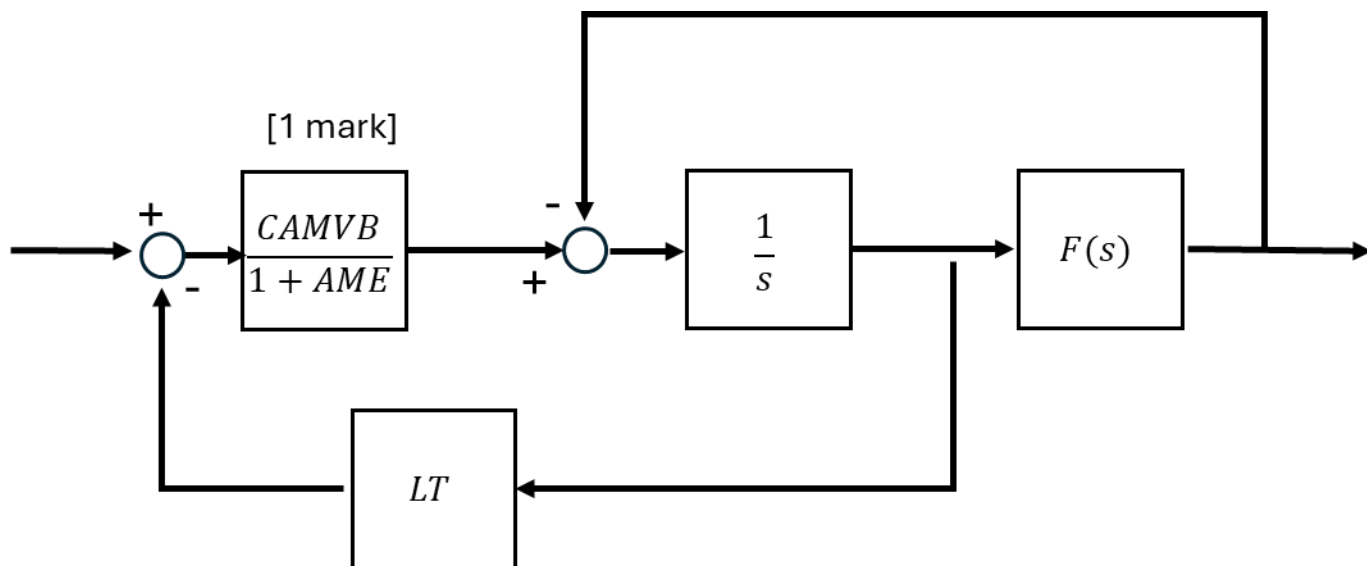
Move the breakaway point of the LT branch to the other side of F (5):



This then simplifies to:

$$\frac{\frac{CAMVBF}{s}}{(1 + AME) \left(1 + \frac{F}{s}\right)} \cdot \frac{CAMVB LT}{1 + \frac{s}{(1 + AME) \left(1 + \frac{F}{s}\right)}}$$

It can be simplified further to create a "neater" transfer function but this is not necessary – we just care about the final cascade and loop simplification. [2 marks]



Question 3 [20 marks]

The African Robotics Unit wants to create a mobile support frame to help its legged robots stay upright during walking and running tests. Think: the walking rings babies use, but for a robot.

Unlike a walking ring, which the baby must push along, the frame will drive itself on wheels and track the position of the robot as it moves, so it doesn't interfere with the robot's motion.

- a) Using the information given in section A of the information sheet on following this question, create a block diagram model of the mobile support frame. Indicate the signals $x(t)$, $u(t)$, $\tau(t)$, $f(t)$, $d_r(t)$ and $d_m(t)$ in your diagram. [10 marks]

- b) Use your block diagram and section B of the information sheet to derive a transfer function for the open-loop system relating the input voltage $u(t)$ to the position $x(t)$ in terms of the given variables. You may assume that the system is initially at rest. [3 marks]

- c) The parameters of the motor and mass of the frame are given in section C of the information sheet, but you do not know the damping factor μ .

To determine μ , you decide to measure the velocity transient response of the frame by removing the motors, pushing it up to some velocity, and then letting it freewheel until it stops moving. The resulting velocity plot is shown in Section D of the information sheet. [4 marks]

- d) Another engineer in the lab thinks that you can simplify your model by representing the motor using only the gain constant k_m . Do you agree? Explain. [3 marks]



Figure 2: baby in a walking ring

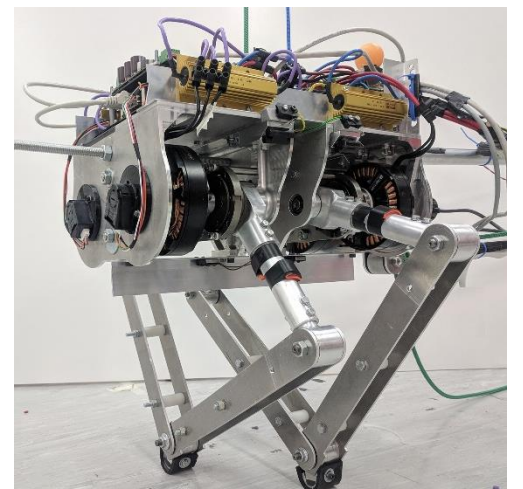


Figure 3: Baleka, the ARU's robot baby (and Gavin's Masters project)

Question 3 – Information

(For convenience, you may remove this sheet by tearing or cutting along the dotted line.)

Section A – System Overview:

- The frame body P is propelled by two driving wheels, each actuated by an electric motor. The motors are controlled by a microprocessor C , which outputs a voltage $u(t)$ to the motors.
- We will assume that the robot and frame are travelling forward in a straight line, so both motors will receive the same input voltage, and therefore produce the same output torque $\tau(t)$.
- Each motor is connected to the wheel by a gearbox that scales the torque up by a factor α .
- The wheel converts the torque into forward force $f(t)$ acting on the frame. The force produced is equal to the applied torque scaled by r , the radius of the wheel.
- The robot is attached to the frame via bungee cables. You can model the effect of the robot and cables as an unknown disturbing force $d_r(t)$ acting on the frame.
- Friction in the gearboxes and wheel mounts opposes the motion of the frame. This force changes based on factors such as temperature, and how well these components have been lubricated. These fluctuations in the model parameters can be captured as another unknown disturbance $d_m(t)$.
- We want the position of the frame to match the position of the robot. The frame measures its distance from the robot $x(t)$ using a proximity sensor, H . The desired value for $x(t)$ is zero.

Section B – Subsystem models

- The frame is a rigid body with mass m acted upon by two forces: the input force (from the wheels and other external sources) and friction. Friction can be modelled as a simple damping effect that is proportional, but opposite to the velocity of the frame $\dot{x}(t)$, with proportionality constant μ .
- The relationship between the voltage u and output torque τ from the motor is approximated by the transfer function $M(s) = \frac{k_m}{\tau_m s + 1}$, where k_m and τ_m are parameters determined by the motor's electronics and dynamics.
- You can assume that sensor H perfectly measures the output position and transmits it instantaneously to the controller.

Section C – Parameters

Parameter	Value
Frame mass m	5
Damping μ	0.3
Gear ratio α	8
Wheel radius r	0.05
Motor gain k_m	0.5
Motor time constant τ_m	0.0012

Section D - Response Tests

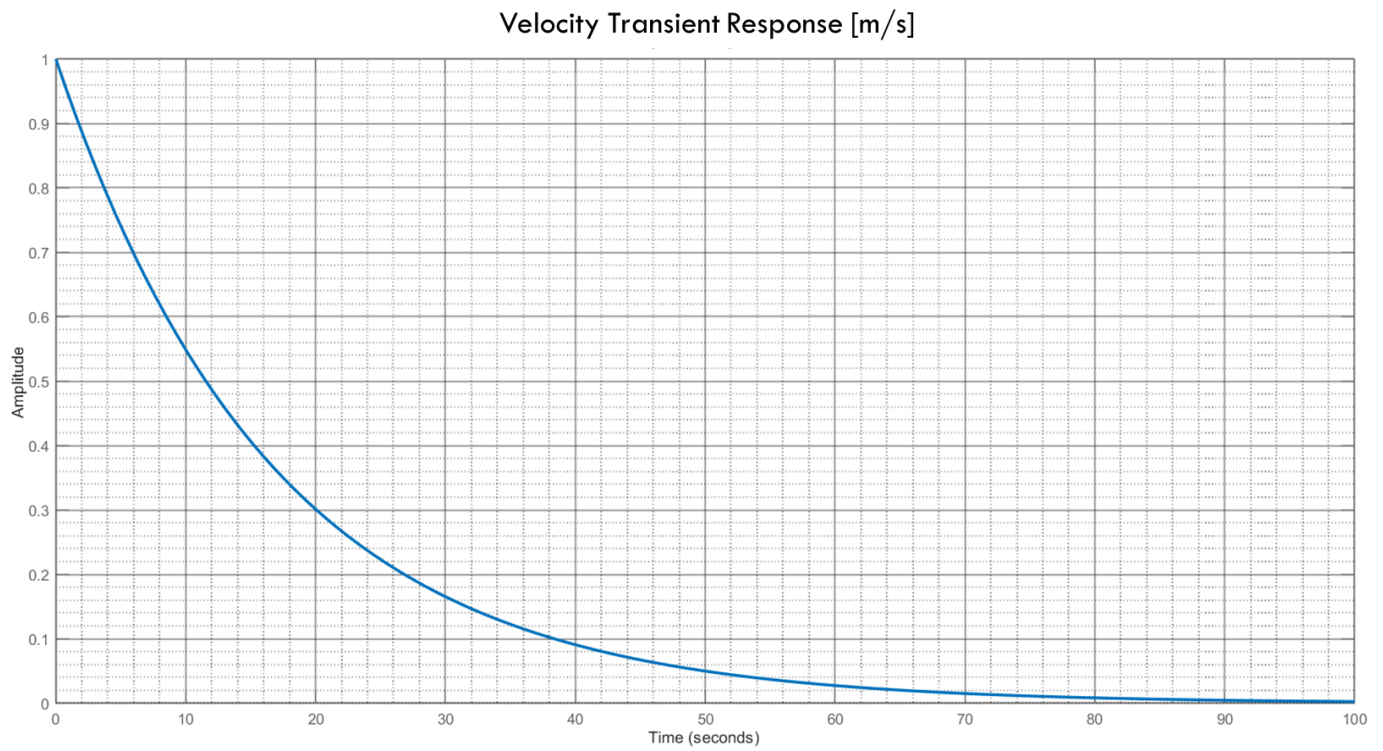
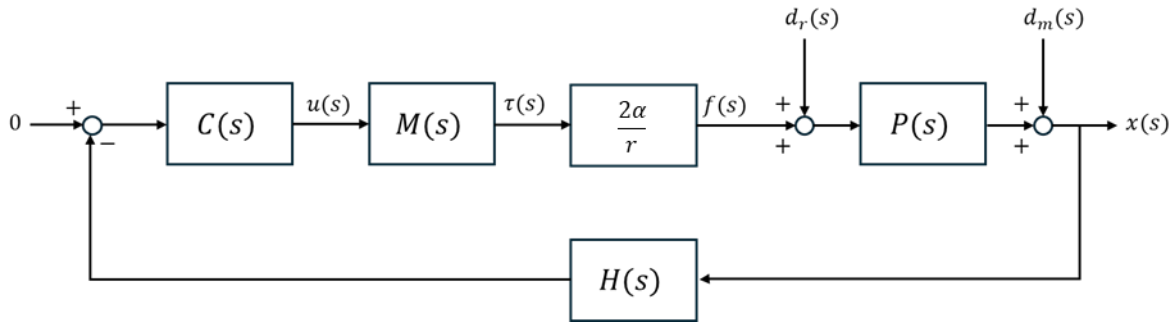


Figure 4: Velocity transient response of support frame with motors removed.

Solution:

a)



[marking:

negative feedback loop structure: 2

controller has correct input and output: 1

actuators (motor, gearbox, wheels) constructed correctly with correct blocks, signals: 3

plant and disturbances are placed and constructed correctly: 3

feedback path is correct: 1

total: 10]

- b) The open-loop model consists of the motors, gearboxes, wheels, and frame body, which are connected in cascade. [1 mark]

The motor transfer function is $M(s) = \frac{k_m}{\tau_m s + 1}$.

The gearbox, wheels and the doubling of the output force to account for both wheels are modelled by the scalar $\frac{2\alpha}{r}$.

The dynamics of the frame body are just a simple first-order damper. This can be represented by the differential equation: $m\ddot{x}(t) = f(t) - \mu\dot{x}(t)$, which has the Laplace transform $\frac{1}{ms + \mu} = \frac{\frac{1}{\mu}}{\frac{1}{\mu}s + 1}$. [1 mark]

This transfer function relates the input force to the velocity, so it can be integrated to give the relationship with position. [1 mark]

This results in the transfer function:

$$G(s) = \frac{2k_m\alpha \frac{1}{r\mu}}{s \left(\frac{m}{\mu}s + 1 \right) (\tau_m s + 1)}$$

- c) The transient response of the system is the same as its impulse response. Without the motors, the transfer function of the system is

$$\frac{G(s)}{M(s)} = \frac{2\alpha \frac{1}{r\mu}}{\frac{m}{\mu}s + 1} = \frac{K}{Ts + 1}$$

so we are going to fit the model of a first-order impulse response, $Ke^{-\frac{1}{T}t}$. The decay rate of the exponential response will then give the time constant, T . [2 marks]

When $t = T$, the response's value will be $e^{-1} = 0.37$ of the initial value. This appears to happen at

around 16 seconds.

Since $T = \frac{m}{\mu} = \frac{5}{\mu} = 16$ we find that $\mu = \frac{5}{16} = 0.3$.

- d) Yes. The time constant of the motor is much faster than the time constant of the frame's dynamics, so the frame's pole will dominate. The motor model can therefore be reduced to its gain.

Question 4 [35 marks]

The support frame is currently being controlled by a proportional controller with a gain of $K = 0.001$. You may use the following transfer function for the open-loop support frame (not including the controller):

$$G(s) = \frac{533}{s(16.7s + 1)}$$

The frequency response of the system and controller is shown in various formats on the information sheet following this question.

- a) Decide whether the following statements about the closed-loop system's tracking ability are TRUE or FALSE. Explain your answers. [4 × 3 marks]
- The frame will be able to match the robot's position perfectly if it is standing still.
 - The frame will be able to match the robot's position perfectly if it is moving at constant velocity.
 - Some sand gets into the wheel shaft, which increases the damping factor μ . This change can be modelled as a constant offset in the output. It does not affect the frame's ability to track the robot's position.
 - One of the frame's non-driving wheels gets stuck, causing it to slide on the ground without rolling. This can be modelled as a constant disturbing force opposing the frame's motion. It does not affect the frame's ability to track the robot's position.
- b) **Transient Response:** We ideally want the frame to reach the robot's position as quickly as possible, with minimal oscillation (less than 10% overshoot) to prevent unnecessary mechanical wear on the system.
- What is the best achievable settling time (to within 5% of the robot's position) that is possible with proportional control? [3 marks]
 - Roughly sketch the root locus of the system, and indicate the region where the closed-loop poles satisfy the given specifications. [6 marks]
 - Determine the range of gains that satisfy the specifications. Does the given controller gain fall into this range? If not, briefly describe how the system will underperform (i.e. too slow, too much overshoot, both). [5 marks]
- c) **Sensitivity:** We want the system to avoid amplifying disturbances by more than a factor of 2.

We are particularly concerned with periodic disturbances caused by the robot's motion. These will have frequencies matching the robot's gait, which will generally have a frequency of around 6 radians per second. These disturbances must not be amplified by more than 2.5%.

- Indicate these specifications on the appropriate plot on your information sheet. [5 marks]
- Does the given controller gain satisfy these specifications? If not, approximate the largest gain that does, and determine the steady-state error with respect to a ramp input at this value (in metres). [4 marks]

Question 4 – Information

(For convenience, you may remove this sheet by tearing or cutting along the dotted line.)

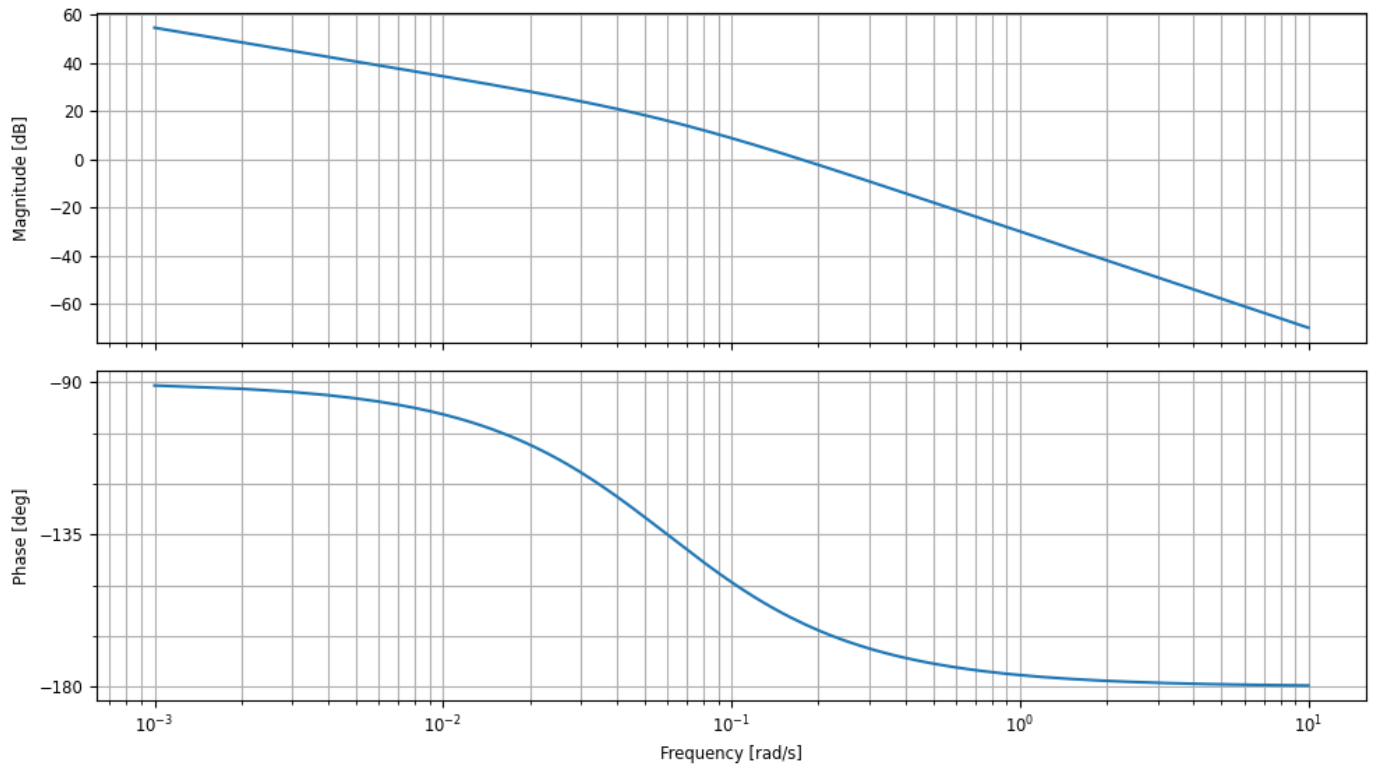


Figure 5: Bode plot of $0.001G(j\omega)$

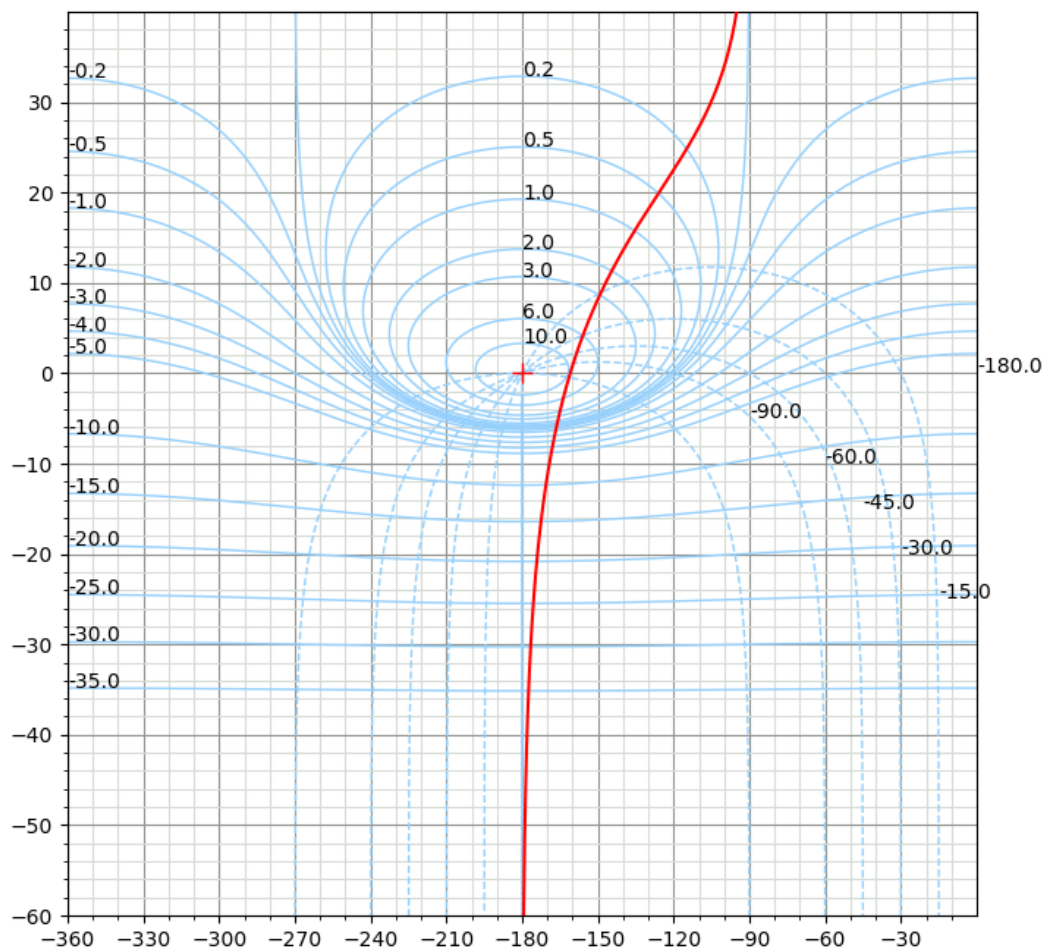


Figure 6: Nichols chart of $0.001G(j\omega)$

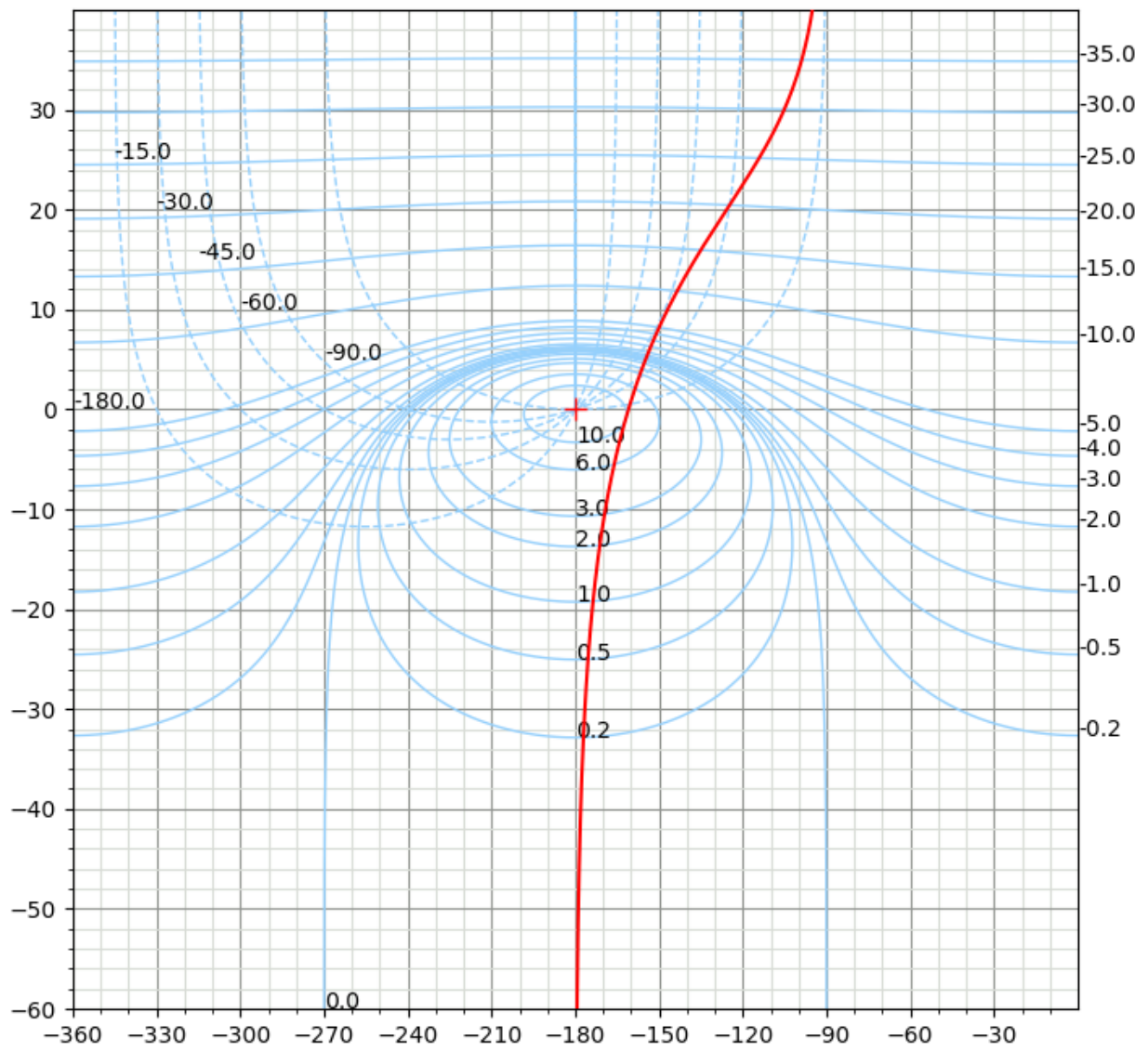


Figure 7: Inverse Nichols chart of $0.001G(j\omega)$

Solutions:

- a) i) TRUE. The system is type 1, so can track position setpoints perfectly.
- ii) FALSE. A type 1 system can't track a ramp input perfectly.
- iii) TRUE. A constant output disturbance will be attenuated completely by a type 1 system.
- iv) FALSE. Although the system is type 1, the proportional controller is type 0. The described situation is a constant *input* disturbance, which specifically requires the controller to have a type number of 1 or greater for it to be rejected.
- b) i) A second-order system is fastest when the poles are on their asymptotes. [1 mark]

The system has poles at 0 and -0.06. The asymptotes will be between these at -0.03. [1 mark]

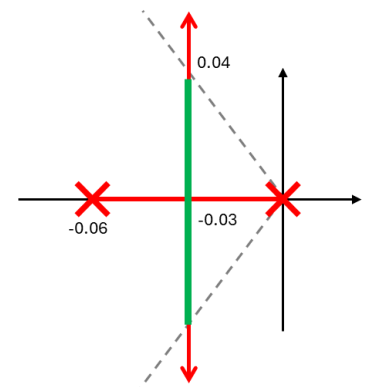
The settling time $\tau_{5\%} = 3\tau = 3 \frac{1}{0.03} = 100$ seconds.

- ii) Root locus with correct specification region [2 marks]

The speed specification requires the poles to be on the asymptotes. The overshoot specification will be satisfied for damping ratios between 1 and 0.6. [1 mark]

When $\zeta = 1$, the system will be critically damped with repeating poles at -0.03. [1 mark]

When $\zeta = 0.6$, the poles must have an angle of $\arccos(0.6) = 53.7^\circ$ with respect to the negative real axis. For a pole with a real part of -0.03, this gives an imaginary part of $\pm 0.03 \tan(53.7^\circ) = \pm 0.041$. [2 marks]



- iii) Minimum gain for critically damped poles: $K = \frac{1}{G(-0.03)} = 2.8 \times 10^{-5}$. [1 mark]

Maximum gain for damping ratio of 0.6: $K = \frac{1}{G(-0.03 + 0.041j)} = 8.1 \times 10^{-5}$. [1 mark]

The proposed gain is larger than this range. [1 mark]

The velocity specification will still be met, as the poles will still be on the asymptotes [1 mark].

The system will, however, have a smaller damping ratio and therefore, more overshoot. [1 mark]

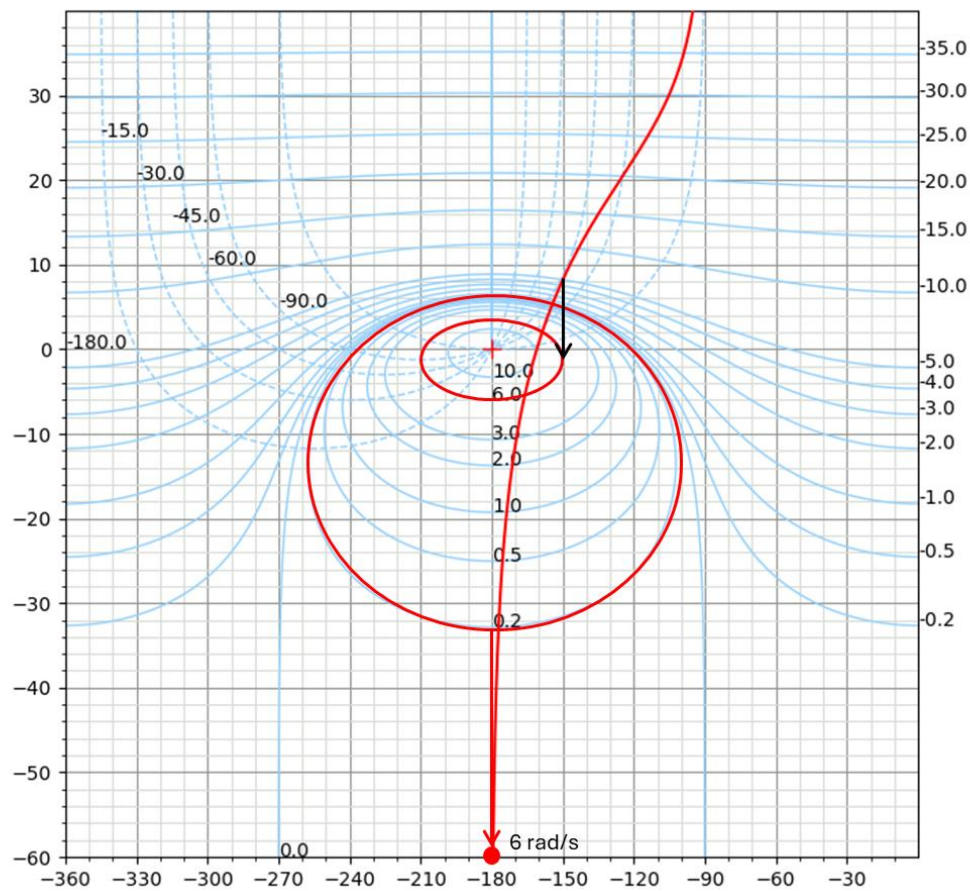
- c) i) The inverse Nichols chart gives information about the sensitivity of the system. [1 mark]

An amplification by a factor of 2 corresponds to sensitivity of magnitude $20 \log_{10}(2) = 6$ dB, so this specification requires that no part of the response crosses into the 60 dB circle. [1 mark]

From the bode plot, we find that, at 6 rad/s, the response has a magnitude of -60 dB and phase of close to -180 degrees, which allows us to locate this frequency on the response. Amplification by 2.5% corresponds to a sensitivity of $20 \log_{10}(1.025) = 0.2$ dB, so this frequency

must stay outside of the 0.2 dB circle. [1 mark]

Specs on plot: [2 marks]



iii) The response passes through the 6 dB circle, so the specifications are violated at this gain [1 mark].

This specification can be satisfied by decreasing the gain by around 9 dB, or 0.35 (black arrow). [1 mark]

This gives a gain of $(0.35)(0.001) = 3.5 \times 10^{-4}$. [1 mark]

At this gain, the steady state error with respect to a ramp is $\lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1+KG(s)} = \frac{1}{KG(0)} = \frac{1}{(3.5 \times 10^{-4})(533)} = 5.3 \text{ m}$ [1 mark]

Question 5 [27 marks]

The proportional controller's performance is not satisfactory, so you decide to design a more sophisticated controller for the support frame.

- a) The main problem with the proportional controller is that the frame does not track the robot accurately enough. Briefly explain why it is not a good idea to try to fix this problem without changing to a different type of controller. [2 marks]

- b) Another student in the lab suggests that the following controller design would be better:

$$C(s) = \frac{1 \times 10^{-6}}{s}$$

The frequency response plots for the system with this controller are given on an information sheet following this question.

- i) There is a major problem with this design. What is it? [2 marks]
- ii) You believe that the problem with this controller can be fixed by adding a lead compensator in cascade with it. Design this lead compensator. Your design should be physically realizable and increase the phase margin of the system to at least 30 degrees. It should not affect the low-frequency gain of the system. [8 marks]
- c) Unfortunately, the additional integrator results in a design that is simply too slow, and the frame is left behind by the robot! Design a new lead compensator from scratch for the original system

$$G(s) = \frac{533}{s(16.7s + 1)}$$

You may wish to support your design with a sketch of the root locus.

Aim to place the dominant poles of the system such that the system has a damping ratio of 0.6 and settles to within 5% of its final value in 1 second. (You do not need to account for uncertainty in your design, you can just choose a target pole with these characteristics.)

Clearly describe your design process, and evaluate whether your system is physically realizable, and able to track a ramp input with ≤ 1 cm (0.01 m) of error. You **do not** need to repeat the design if one of these requirements is not met, but you should briefly explain how the design could be changed to do so. [15 marks]

Question 5 – Information

(For convenience, you may remove this sheet by tearing or cutting along the dotted line.)

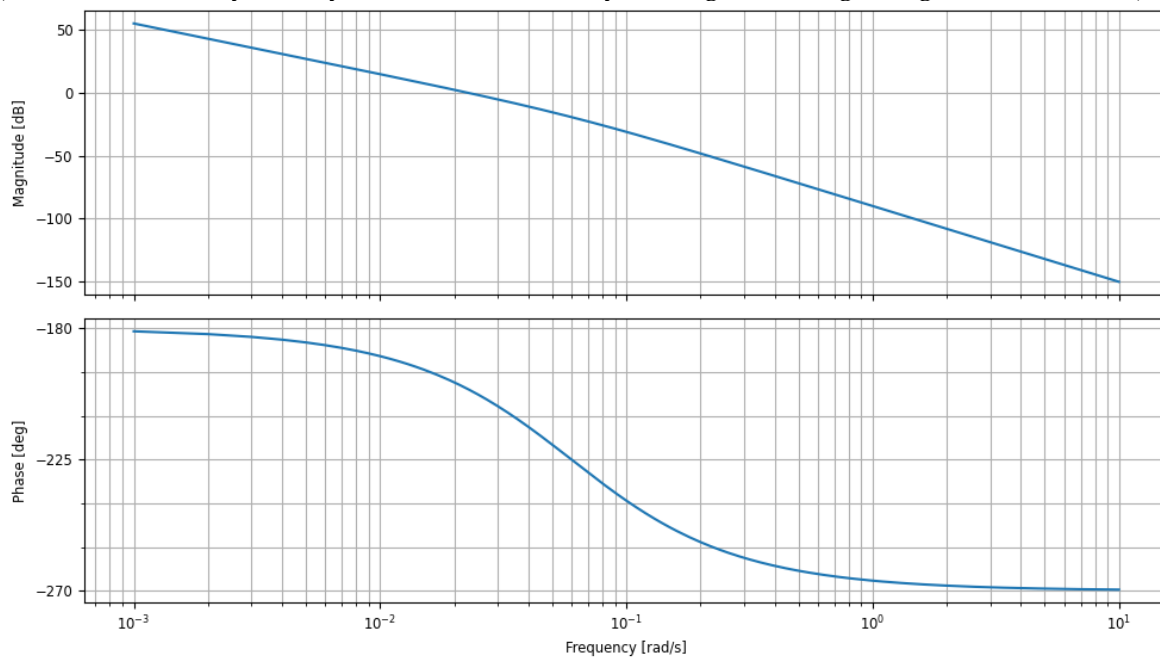


Figure 8: Bode plot of $C(s)G(s)$.

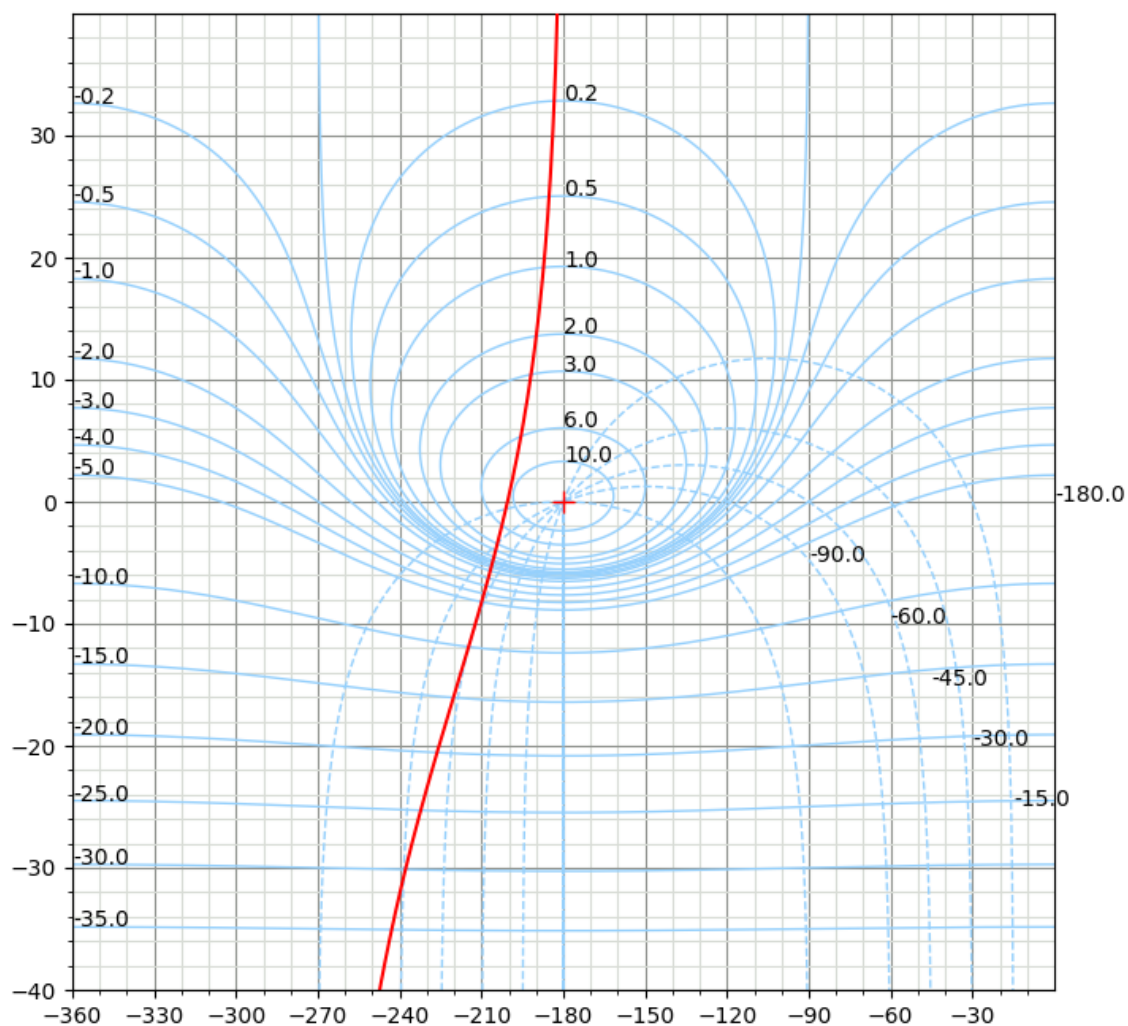


Figure 9: Nichols chart of $C(s)G(s)$

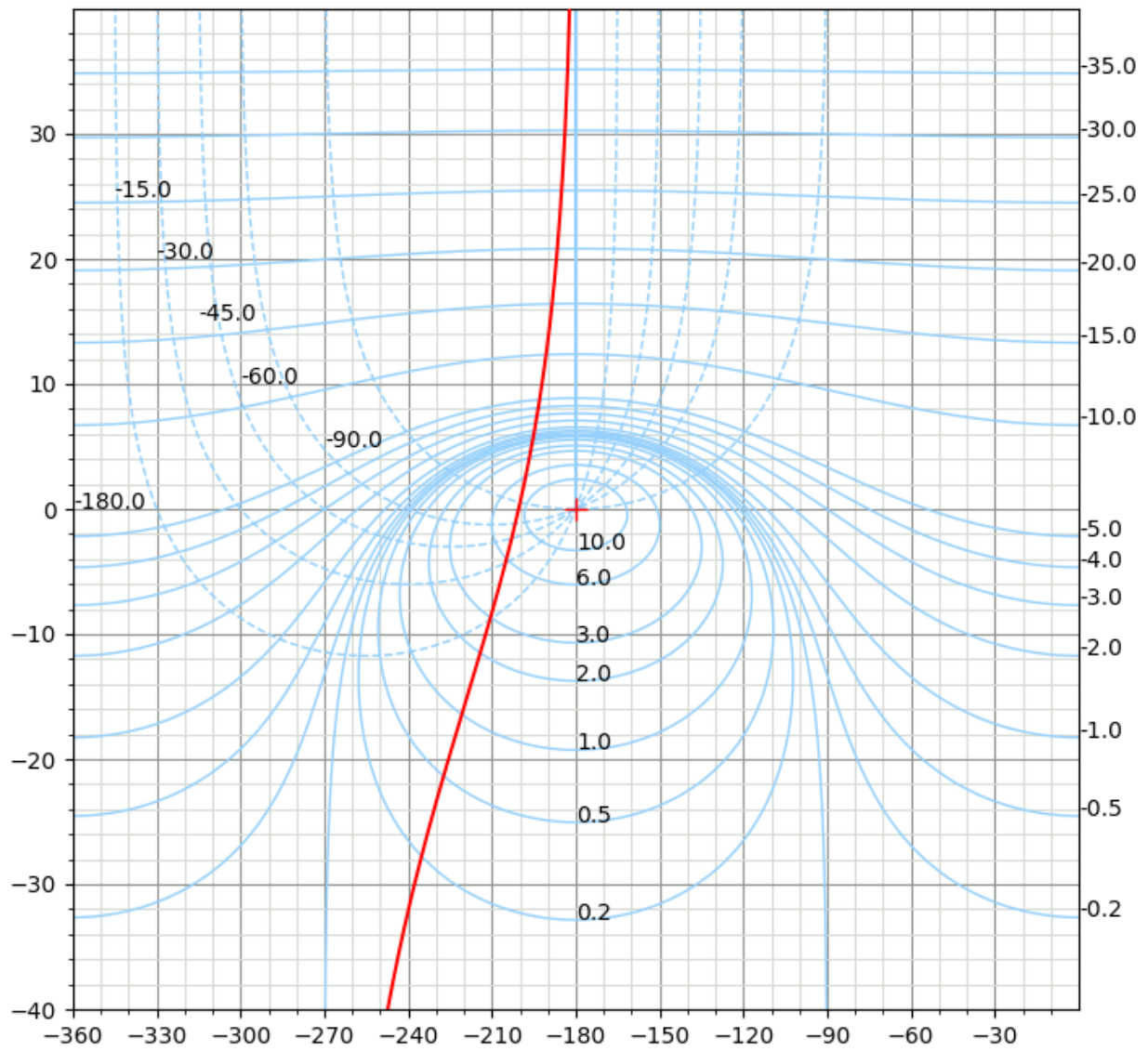


Figure 10: Inverse Nichols chart of $C(s)G(s)$

Solutions:

- a) Improving the tracking would require increased low-frequency gain. Increasing the gain of the proportional controller would increase the gain at all frequencies, which would make the system oscillatory and sensitive. [2 marks]
- b) i) The closed-loop system will be unstable at all gains. This can be seen on the Nichols chart, where the response encircles -1, or by sketching a root locus, where the two poles at the origin immediately become unstable. [2 marks]
- ii) A lead compensator has the form $\frac{\alpha\tau s + 1}{\tau s + 1}$.

The system's phase at the gain crossover is currently around -20 degrees, so it must be increased by a total of 50 degrees to stabilize the system with the required phase margin. [1 mark]

Using the formula $\sin(\Phi_p) = \frac{\alpha - 1}{\alpha + 1}$, we find that the required α is 7.5. [1 mark]

It is advisable to increase this, so we will go with $\alpha = 10$. (This is arbitrary so the marks following this are based only on the procedure for whatever value is selected.) [1 mark]

The gain at the compensator peak will be $10 \log_{10}(\alpha) = 10$ dB, [1 mark]
so we must place the midpoint frequency ω_m at the value where the gain is -10 dB, so that this will become the new gain crossover when the responses of the controller and plant are combined.

From the bode plot, we can eyeball that the frequency where the gain is -10 dB is around 0.04 rad/s. [1 mark]

Using $\omega_m = \frac{1}{\tau\sqrt{\alpha}} \therefore 0.04 = \frac{1}{\tau\sqrt{10}}$ we find $\tau = \frac{1}{0.04\sqrt{10}} = 7.9$. [1 mark]

This gives a compensator $\frac{7.9s + 1}{7.9s + 1}$. [1 mark]

Design is physically realizable (pole and zero separated by one order of magnitude) and formatted to have a gain of 1 [1 mark].

- c) Desired CL pole: (2)

For $\tau_{5\%} \leq 1, 3\tau \leq 1 \therefore \tau \leq \frac{1}{3}$ so the pole location must be ≤ -3 . [1 mark]

For a damping ratio of 0.6, the angle of the pole to the negative real axis must be $\arccos(0.6) = 53.1^\circ$, so the imaginary part is $3 \tan(53.1^\circ) = 4$. [1 mark]

So, we will aim for a dominant closed loop pole pair at $-3 \pm 4j$.

Angle criterion: (2)

The existing poles in the system have a combined angle of -253.2° with respect to the desired pole. This can be found through trigonometry, or obtaining $\angle G(s_d)$. [1 mark]

The compensator must add $\theta_c = 73.2^\circ$ to reach the closest odd multiple of 180 degrees [1 mark].

Zero placement: (3)

We want to avoid placing the zero further right than the desired pole, so the upper limit on its position is -3. [1 mark]

The zero will contribute θ_c by itself at $-3 - \frac{4}{\tan(73.2^\circ)} = -4.2$, so the zero should not be placed left of this point. [1 mark]

Let's try a zero at -3. [1 mark]

(This is an arbitrary choice so marks from here will consider the procedure rather than specific values.)

Pole placement (2):

The zero contributes an angle of $\theta_z = 90^\circ$, and $\theta_c = \theta_z - \theta_p$, so the pole's angle contribution must be $\theta_p = -16.8^\circ$. [1 mark]

The pole will have the needed angle at $-3 - \frac{4}{\tan(-\theta_p)} = -16.2$. (Note the reversal of the angle, because the pole's angle is subtracted while the zero's is added.) [1 mark]

Gain selection (2):

The gain-normalized compensator is $C(s) = \frac{\frac{1}{z}s+1}{\frac{1}{p}s+1} = \frac{0.33s+1}{0.06s+1}$. [1 mark]

The required gain is $K = \frac{1}{|C(s_d)G(s_d)|} = 0.5$ [1 mark]

Evaluation (4):

The compensator is physically realizable because the pole is less than 10 times further from the origin than the zero is. [1 mark]

(If the pole was too far away, trying a zero closer to the origin would be the recommendation.) [1 mark]

The final value of the error for a ramp is $\lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1+KC(s)G(s)} = \frac{1}{KC(0)G(0)} = \frac{1}{(0.5)(533)} = 0.038$.

This is less than 1 cm (0.01) so it satisfies the accuracy requirement. [1 mark]

(If it did not, the gain could be increased by moving the zero further away, if that is possible within the constraints of physical realizability.) [1 mark]

Extra Space

[illegible]

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} \right)}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>